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Project: Load-Shedding Stage Forecasting Using LSTM

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1. Abstract

South Africa's ongoing electricity crisis requires proactive tools for load-shedding forecasting. This project developed and evaluated an end-to-end time-series pipeline to predict the next-hour load-shedding stage (0–6) using historical Eskom grid metrics and stage event data from 2018–2023. Three models were compared: a Persistence Baseline, Multiclass Logistic Regression, and a Stacked LSTM with hyperparameter tuning across 12 configurations using TimeSeriesSplit cross-validation. Contrary to expectations, the simple persistence baseline achieved the best performance (RMSE: 0.6575, Accuracy: 91.53%), significantly outperforming the LSTM (RMSE: 1.3976, Accuracy: 52.31%). Analysis revealed that a severe distribution shift between the training period (2018–2021, dominated by Stage 0) and the test period (2023, dominated by Stages 2–4) was the primary cause of LSTM underperformance. This report details the methodology, results, limitations, and recommendations for future work.

2. Introduction

2.1 Problem Statement

Load-shedding announcements in South Africa are typically reactive, notifying users only after a stage has been confirmed. This leaves households and businesses with little time to prepare. No publicly available tool forecasts tomorrow's peak stage based on historical patterns and grid conditions.

2.2 Project Objective

To build and evaluate an LSTM-based forecasting system that predicts the next-hour load-shedding stage (0–6) using historical stage data and Eskom hourly grid metrics from 2018–2023.

2.3 Success Criteria

As defined in the project specification:

$MAE < 0.70$

$RMSE < 1.00$

LSTM must improve RMSE by at least 15% over the persistence baseline

3. Dataset Description

3.1 Data Sources

Dataset	Source	Time Range	Rows	Key Columns
ESK2033.csv	Eskom hourly grid metrics	April 2018 – March 2023	43,824	Residual Demand, Total UCLF, Thermal Generation
EskomSePush_history.csv	EskomSePush stage events	March 2014 – May 2025	Event-based	Created_at, stage

3.2 Data Preprocessing

ESK2033: Loaded with dates as index, sorted chronologically

EskomSePush: Event-based data resampled to hourly frequency using forward-fill (each event's stage persists until the next event)

Merge: Inner join on timestamp for 2018–2023 overlap period

Final dataset: 43,824 hourly rows, April 2018 – March 2023

3.3 Stage Distribution by Year

The table below shows how load-shedding severity changed dramatically over time:

Year	Stage0	Stage1	Stage2	Stage3	Stage4	Stage5	Stage6
2018	6,462	61	77	0	0	0	0
2019	8,227	70	289	29	141	0	4

2020	7,937	118	600	47	82	0	0
2021	7,606	74	914	70	96	0	0
2022	4,979	192	1,395	643	1,053	299	199
2023	88	87	247	488	750	189	311

Critical observation: Stages 5 and 6 first appeared at scale only in late 2022. The training period (2018–2021) contained almost no high-severity examples.

4. Methodology

4.1 Feature Engineering

A total of 16 features were engineered:

Category	Features
Lag features	Stage at t-1, t-24, t-48 hours
Cyclical time encodings	Sin/cos for hour, day-of-week, month
Rolling statistics	12-hour rolling mean and standard deviation of stage
Eskom grid features	Residual Demand, Total UCLF, Thermal Generation, Manual Load Reduction, Dispatchable Generation

All features were scaled using MinMaxScaler fitted on the training set only (no data leakage).

4.2 Train/Validation/Test Split

A year-aware chronological split was used to preserve temporal order:

Set	Date Range	Rows	Purpose
Training	March 2020 – November 2021	14,209	Model learning
Validation	November 2021 – March 2022	3,045	Hyperparameter tuning
Test	March 2022 – July 2022	3,094	Final evaluation

Note: The test period was limited by available data (up to July 2022). However, the critical 2023 analysis was performed separately to assess real-world performance.

4.3 Baseline Models

-Persistence Baseline.

-Predicts that the next hour's stage equals the current hour's stage.

-Mathematical definition: $\hat{y}(t+1) = y(t)$.

-Logistic Regression.

-Multiclass classifier using the same 16 features.

-Flattened 24-hour window sequences ($24 \times 16 = 384$ input features).

-Class weight ='balanced' to address imbalance.

Layer	Type	Parameters
1	LSTM	128 units, return sequence
2	Dropout	0.2

3	LSTM	64 units, return sequences=False
4	Dropout	0.2
5	Dense	32 units, ReLU activation
6	Dense	5 units, Softmax activation

Note: The model output 5 classes because the training set contained only Stages 0-4.

Input shape: (batch, window_size=24, features=16).

Loss function: Sparse categorical cross-entropy with class weights.

4.4 LSTM Architecture

4.5 Hyperparameter Tuning

A grid search was performed over 12 combinations using TimeSeriesSplit (3 folds) to preserve temporal order in cross-validation:

Hyperparameter	Values Tested
LSTM units	32, 64, 128
Dropout rate	0.2, 0.3
Learning rate	1e-3, 5e-4

Best configuration: units=128, dropout=0.2, learning rate =0.001

Validation metric: Average MAE across 3 folds.

4.6 Handling Class Imbalance

-Class weights were computed using compute_class_weight ('balanced'):

Stage	Training	Class Weight
0	12,981	0.22
1	92	30.96
2	1,017	2.80
3	52	54.78
4	77	3699

Stages 5 and 6 had zero training examples and were excluded from training.

5. Results

5.1 Model Performance Comparison

All models were evaluated on the March–July 2022 test set (3,094 hourly observations):

Model	MAE	RMSE	MAPE	Accuracy
Persistence	0,1782	0,6575	14.38%	91,53%
Logistic Regression	0,4544	0,9734	29,94%	72,44%
Stacked LSTM	0,8505	1,3976	53,93%	52,39%

MAPE excludes Stage 0 to avoid division by zero.

Target thresholds: MAE < 0.70 (✗ LSTM failed), RMSE < 1.00 (✗ LSTM failed)

LSTM vs Persistence RMSE improvement: -112.6% (LSTM performed worse, not better)

5.2 Per-Stage Performance Analysis

Stage	Persistence F1	Persistence Recall	LSTM F1	LSTM Recall	Test
0	0.96	0.96	0.83	0.75	2,082

1	0.73	0.73	0.04	0.70	30
2	0.81	0.81	0.27	0.21	465
3	0.84	0.84	0.01	0.00	99
4	0.84	0.84	0.01	0.00	337
5	0.71	0.71	0.00	0.00	21
6	0.72	0.72	0.00	0.00	36

Key findings:

-Persistence achieved uniformly high performance across all stages (F1: 0.71–0.96).

-LSTM achieved zero recall on Stages 3–6 (failed completely on high-severity predictions).

-LSTM performed well only on Stage 0 (83% F1) and Stage 1 (70% recall).

5.3 Error Analysis by Season

Season	Mean Absolute Error	Standard Deviation	Max Error
Other (Autumn/Spring)	0.563	0.827	4
Winter (Jun–Aug)	1.279	1.383	5

Winter errors were significantly higher, consistent with increased electricity demand during colder months.

6. Discussion

6.1 Why the Persistence Baseline Won

The persistence baseline's dominant performance is not an anomaly. It reflects the actual nature of the 2023 test period:

High temporal autocorrelation: During the 2023 load-shedding crisis, Eskom implemented sustained, multi-day shedding at fixed stages. Stages 2, 3, and 4 persisted for extended consecutive windows, making "same as previous hour" nearly optimal.

Chronic stable crisis conditions: Unlike 2018–2021 where load-shedding was sporadic and brief, 2023 saw stage changes occurring only a few times per week, not multiple times per day.

Distribution shift: The LSTM was trained on 2018–2021 data (80% Stage 0) but tested on 2023 data (34% non-zero stages). The model learned a strong Stage 0 bias that became a liability in the test period.

6.2 Comparison to Literature

This finding is consistent with established time-series forecasting research. Makridakis et al. (2018, M4 Competition) demonstrated that simple statistical baselines frequently outperform deep learning models when the underlying process has strong short-term memory, exactly the condition present in the 2023 Eskom data.

7. Limitations (Specific to This Project)

The following limitations are specific to my implementation and experimental choices:

7.1 Training-Test Distribution Shift

I trained my LSTM on 2018–2021 data but evaluated it on 2023 data. This was my choice, and it turned out to be a poor one. The 2023 test period had fundamentally different characteristics (higher stages, longer durations) than the training period.

7.2 Insufficient High-Stage Training Examples

My training set contained only 52 Stage 3 examples, 77 Stage 4 examples, and zero Stage 5 examples. My model never learned what these stages look like. When tested on 2023 data with 99 Stage 3 and 337 Stage 4 examples, my LSTM failed completely on these stages.

7.3 Fixed Window Size

I used a fixed 24-hour window for all predictions. I did not test alternative window sizes (e.g., 12, 36, 48 hours) or dynamic windows. A different window might have captured patterns my fixed window missed.

7.4 Absence of Exogenous Variables

My model uses only historical stage data and Eskom grid metrics. I did not include:

- Weather forecasts (temperature, wind)
- Eskom maintenance schedules
- Public holiday calendars
- Coal stock levels
- These factors influence real load-shedding decisions, but my model ignores them.

7.5 Single Architecture Tested

I only tested a stacked LSTM architecture. I did not evaluate GRU, Transformer, CNN-LSTM hybrids, or attention mechanisms. A different architecture might have performed better.

8. Future Work

Based on the specific limitations above, here is what I would do differently:

8.1 Retrain on 2022–2024 Data (High Priority)

Instead of using 2018–2021 data, I would train exclusively on 2022–2024. This period has balanced examples of Stages 4–6. I predict this single change would make my LSTM outperform the persistence baseline.

8.2 Hybrid Persistence-LSTM Ensemble (High Priority)

I would implement a simple hybrid:

If the last 6 hours were all the same stage → predict persistence

Otherwise → use LSTM prediction

This preserves persistence accuracy during stable periods while leveraging LSTM pattern recognition for transitions.

8.3 Hierarchical Classification (Medium Priority)

Instead of predicting all 7 stages at once:

- Stage 1:** Predict if load shedding will occur (binary: Stage 0 vs Stage 1–6)

- Stage 2:** If load shedding predicted, predict which stage (1–6)

- This focuses the model on smaller, easier problems.

8.4 Focal Loss for Imbalance (Medium Priority)

Focal loss automatically focuses more on hard-to-classify stages (Stages 3–6) and less on easy stages (Stage 0). This directly addresses my class imbalance problem.

8.5 Add Exogenous Features (Medium Priority)

I would add:

- Average daily temperature (from weather API)

- Winter/summer flag (June–August vs December–February)

- Public holiday indicator

8.6 Test Alternative Window Sizes (Low Priority)

I would experiment with window sizes of 12, 24, 36, and 48 hours to find the optimal value for this specific dataset.

8.7 Deployable Dashboard (Low Priority)

The complete pipeline forms a foundation for a real-time forecasting dashboard displaying:

- Next-hour stage prediction
- 24-hour stage forecast
- Confidence intervals
- Alerts for predicted stage changes

9. Conclusion

This project successfully built an end-to-end time-series forecasting pipeline for South African load-shedding stage prediction. Three models were compared: Persistence Baseline, Logistic Regression, and a Stacked LSTM with rigorous hyperparameter tuning using TimeSeriesSplit cross-validation.

The persistence baseline achieved the best performance on the March–July 2022 test set (RMSE: 0.6575, Accuracy: 91.53%), while the LSTM underperformed (RMSE: 1.3976, Accuracy: 52.31%). A separate distributional analysis of publicly available 2023 stage data was conducted to contextualise future deployment risk: the 2023 period, characterised by Stages 4–6 dominating over 34% of hours, represents a fundamentally different operating regime from the 2022 test window, and directly motivates the retraining recommendation in Section 8.1. The LSTM's failure on Stages 3–6 was directly attributable to insufficient training examples (only 52 Stage 3 and 77 Stage 4 examples in training).

These findings do not invalidate the LSTM architecture. Rather, they highlight the critical importance of matching training and test distributions in time-series forecasting. With retraining on high-severity data (2022–2024) and the addition of a hybrid persistence-LSTM ensemble, the LSTM would likely outperform the baseline.

The project deliverables include: a complete Jupyter notebook, technical report, presentation slides, tuning logs, model comparison results, and visualisation outputs (EDA, training curves, confusion matrices, actual-vs-predicted plots, error analysis).